

Week 10 Implementation Report: RAG Pipeline Engineering

Project: LearnPulse AI LMS | **Component:** RAG & Vector Search | **Date:** Oct 1, 2026 | **Status:** Implemented & Verified

1. Executive Summary & Objective

Week 10 implements the complete end-to-end Retrieval-Augmented Generation (RAG) pipeline for the LearnPulse platform. The RAG architecture connects existing document extraction tools with token-aware chunking, vector embedding generation, vector storage in PostgreSQL via pgvector, similarity retrieval, document-grounded prompt engineering, and Ollama-hosted local LLMs.

2. Approved Technology Stack & Configuration

Component	Technology / Value	Description / Responsibilities
Embedding Model	nomic-embed-text:latest	Dedicated text embedding model hosted via Ollama.
Embedding Dimension	768	Strictly validated vector size stored in pgvector.
Generative LLM	qwen2.5-coder:7b	Existing local chat model used exclusively for final answer generation.
Vector Store	PostgreSQL 16 + pgvector	Table vector_store using COSINE distance and HNSW index.
Text Splitter	TokenTextSplitter	Spring AI token-aware splitter maintaining chunk metadata.
Chunk Size / Overlap	800 / 100 tokens	Configurable via application.yml (min-size=20).
Top-K / Threshold	Top-K = 4 / Threshold = 0.5	Configurable similarity retrieval parameters.

3. Architecture & Implementation Highlights

- **DocumentChunkingService:** Token-aware chunking preserving documentId, fileName, uploadedBy, and deterministic chunkIndex.
- **DocumentEmbeddingService:** Wraps Spring AI EmbeddingClient with nomic-embed-text, enforcing 768-dimension output validation.
- **PgVectorStore Configuration:** Configures Spring AI PgVectorStore with 768 dimensions and COSINE distance matching.
- **DocumentRetrievalService:** Top-K=4 similarity search with strict documentId metadata filtering.
- **Idempotent Ingestion & No-Context Rule:** Prevents duplicate vector ingestion. If no chunks exceed threshold=0.5, returns explicit 'could not be found' message without calling LLM.
- **API Boundary:** POST /api/ai/ask-document restricted to ROLE_STUDENT via JWT.

4. Automated Testing & Regression Results

The complete automated Maven test suite was executed against OpenJDK 21.

Total Suite Execution Time: 47.869 seconds | **Total Tests Executed:** 115 | **Passed:** 115 | **Failures:** 0 | **Errors:** 0 |

Skipped: 0

Build Result: BUILD SUCCESS

- **New Week 10 Tests (21 New):** DocumentChunkingTest, DocumentEmbeddingTest, DocumentRetrievalIntegrationTest, DocumentRagIntegrationTest.
- **Regression Verification:** All 94 existing Week 1–9 tests passed with zero regressions.

5. Real End-to-End Verification (verify_week10_rag.py)

Validated against live Spring Boot application, PostgreSQL 16 + pgvector, and Ollama using sample PDF Java_Programming_Fundamentals.pdf (3,192 chars, Document ID: f863c45a-be29-45dc-8b77-44af6aadb5a).

Query	Status	Latency	Score	Answer / Result Summary
What is inheritance?	HTTP 200	38,281 ms	0.3292	Grounded answer detailing extends, superclass, subclass, single inheritance, and @Override.
What is polymorphism?	HTTP 200	27,311 ms	0.4175	Grounded answer detailing overloading vs overriding and dynamic method dispatch.
Explain interfaces.	HTTP 200	23,633 ms	0.4156	Grounded answer detailing abstract methods, static constants, default & private methods.
What is exception handling?	HTTP 200	26,435 ms	0.4077	Grounded answer detailing Throwable, checked/unchecked exceptions, try-catch-finally.
What is the capital of France?	HTTP 200	234.21 ms	0 sources	Explicit no-context notice: 'The requested information could not be found in the selected document.'

6. Security Verification & Scope Boundaries

- **Unauthenticated Access:** POST /api/ai/ask-document without JWT token returned **HTTP 401 Unauthorized**.
- **Invalid Document UUID:** Querying missing UUID returned **HTTP 404 Not Found**.
- **Scope Boundary:** Backend RAG pipeline strictly implemented; no frontend or external paid APIs added.
- **Git Confirmation:** No Git commits or pushes were performed; working tree remains preserved.

Final Status: Implemented and verified against the Week 10 specification.